Relations
Lecture N°5
1 Lic Data Science

DJOUMAKH akila

Faculty of Mathematics
ALGEBRA1

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Relations
Definitions
Properties of relations
Relations d'équivalence
Equivalence classes and quotient set
Partition
Orders on a set

Summary

Relations

Definitions

- Properties of relations
- Relations d'équivalence
  - Equivalence classes and quotient set
  - Partition
- Orders on a set

Binary relation

Definition

A binary relation $\mathcal{R}$ on sets E and F is a definite relation between elements of E and elements of F. We write $x\mathcal{R}y$ if $x \in E$ and $y \in F$ are related. If $E = F$, then we call $\mathcal{R}$ a relation on E.

Definition

Let E be a non-empty set. A **binary relation** (or simply a relation) $\mathcal{R}$ on E is any subset of $E \times E$. One usually writes $x\mathcal{R}y$ when $(x, y) \in \Gamma$, and we say that x is in relation with y. Γ is called **the graph** of $\mathcal{R}$.

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Example

- Suppose that S is a set of students enrolled in a university and B is a set of books in a library. We might define a relation $\mathcal{R}$ on S and B by :
  $s \in S$ has read $b \in B$. In that case, sRb if and only if s has read b. Another, probably inequivalent relation is :
  $s \in S$ has checked $b \in B$ out of the library.
- For $E = \mathbb{N}$ consider the property $\mathcal{R}_2(x, y)$ defined by :
  (x, y) checks the property $\mathcal{R}_2$ if x divides y, this means that there exists $k \in \mathbb{N}$ such that $y = kx$. Thus, we have $\mathcal{R}_2(0, 0)$ and $\mathcal{R}_2(2, 0)$, but we do not have $\mathcal{R}_2(0, 2)$
- The subset Γ_1 defined by $\Gamma_1 = \{(1, 2), (1, 3), (2, 1)\}$ is a binary relation on $S = \{1, 2, 3\}$.
- The subset Γ_2 of $\mathbb{R}^2$, defined by
  $\Gamma_2 = \{(x, y) \in \mathbb{R}^2 : x < y\}$ is a binary relation on $\mathbb{R}$.

Type of relations: reflexivity, symmetry, transitivity

Definition

Let $\mathcal{R}$ be a binary relation on E,

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Type of relations: reflexivity, symmetry, transitivity

Definition

Let $\mathcal{R}$ be a binary relation on E,

1. $\mathcal{R}$ is called **reflexive** relation if :$\forall x \in E, \quad x\mathcal{R}x$.

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Type of relations: reflexivity, symmetry, transitivity

Definition

Let $\mathcal{R}$ be a binary relation on E,

1. $\mathcal{R}$ is called **reflexive** relation if :$\forall x \in E, \quad x\mathcal{R}x$.
2. $\mathcal{R}$ is called **symmetric** relation if:
   $\forall x, y \in E, \quad x\mathcal{R}y \Rightarrow y\mathcal{R}x$.

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Type of relations: reflexivity, symmetry, transitivity

Definition

Let $\mathcal{R}$ be a binary relation on E,

1. $\mathcal{R}$ is called **reflexive** relation if :$\forall x \in E, \quad x\mathcal{R}x$.
2. $\mathcal{R}$ is called **symmetric** relation if:
   $\forall x, y \in E, \quad x\mathcal{R}y \Rightarrow y\mathcal{R}x$.
3. $\mathcal{R}$ is called **transitive** relation if:

$$\forall x, y, z \in E, \quad (x\mathcal{R}y \quad \textit{and} \quad y\mathcal{R}z) \Rightarrow x\mathcal{R}z.$$

Type of relations: reflexivity, symmetry, transitivity

Definition

Let $\mathcal{R}$ be a binary relation on E,

1. $\mathcal{R}$ is called **reflexive** relation if :$\forall x \in E, \quad x\mathcal{R}x$.
2. $\mathcal{R}$ is called **symmetric** relation if:
   $\forall x, y \in E, \quad x\mathcal{R}y \Rightarrow y\mathcal{R}x$.
3. $\mathcal{R}$ is called **transitive** relation if:

$$\forall x, y, z \in E, \quad (x\mathcal{R}y \quad \textit{and} \quad y\mathcal{R}z) \Rightarrow x\mathcal{R}z.$$

4. $\mathcal{R}$ is called antisymmetric relation if :
   $\forall x, y \in E, \quad (x\mathcal{R}y \quad \textit{and} \quad y\mathcal{R}x) \Rightarrow x = y$.

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Example

Take $m \in \mathbb{Z}_+$ and let a, b, and c be integers. Define a to be **congruent** to b modulo m and write

$$a \equiv b(mod m)/ \text{ if and only if } m|a-b$$

That is,

$$a \equiv b(mod m) \text{ if and only if } a = b + mk \text{ for some } k \in \mathbb{Z}.$$

For example, we have that $7 \equiv 1(mod 3)$, $1 \equiv 13(mod 3)$, and $27 \equiv 0(mod 3)$, Congruence modulo m defines the relation

$$R_m = \{(a, b) : a \equiv b(mod m)\}$$

Example

Observe that

$$R_m = \{(a,b) : m|a-b\} = \{(a,b) : \exists k \in \mathbb{Z} \wedge a = b + mk\}$$

Prove that R_m is an equivalence relation.

- $(a,a) \in R_m$ because $a \equiv a(mod m)$.
- Assume $(a,b) \in R_m$. This implies that $m|a-b$.then $m|b-a$. Hence, $(b,a) \in R_m$.
- Let $(a,b),(b,c) \in R_m$. Then $a = b + mk$ and $b = c + ml$ for some $k, l \in \mathbb{Z}$. Substitution yields

$$a = c + ml + mk = c + m(l+k).$$

Since the sum of two integers is an integer, $(a,c) \in R_m$.

Definition

Let $\mathcal{R}$ be a relation on a set A. The relation $\mathcal{R}$ is said to be an **an equivalence** relation if it satisfies the following conditions:

- $\mathcal{R}$ is reflexive,
- $\mathcal{R}$ is antisymmetric,
- $\mathcal{R}$ is transitive.

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Example

Define a relation $\mathcal{R}$ on $\mathbb{Z}$ by $x\mathcal{R}y$ if x and y have the same parity (even or odd). We claim that $\mathcal{R}$ is an equivalence relation:

1. Reflexivity: Since x has the same parity as x, $x\mathcal{R}x$.
2. Symmetry: If $x\mathcal{R}y$, then x and y have the same parity. Thus y and x have the same parity, and hence $y\mathcal{R}x$.
3. Transitivity: If $x\mathcal{R}y$, then x and y have the same parity. If $y\mathcal{R}z$, then y and z have the same parity. Since y has only one parity, we can thus conclude that x and z have the same parity, so $x\mathcal{R}z$.

Therefore $\mathcal{R}$ is an equivalence relation.

Example

Let R be the relation on the set $\mathbb{R}$ real numbers defined by xRy *iff* $x-y$ *is an integer*. Prove that R is an equivalence relation on $\mathbb{R}$.

Proof:

- Reflexive: Suppose $x \in \mathbb{R}$. Then $x - x = 0 \in \mathbb{Z}$, which is an integer. Thus, xRx.
- Symmetric: Suppose $x, y \in \mathbb{R}$ and xRy. Then $x - y$ is an integer. Since $y - x = -(x - y)$,$y - x$ is also an integer. Thus, yRx.
- Suppose $x, y, z \in \mathbb{R}$, xRy and yRz. Then $x - y$ and $y - z$ are integers. Thus, the sum $(x - y) + (y - z) = x - z$ is also an integer, and so xRz. Thus, R is an equivalence relation on $\mathbb{R}$.

Equivalence classes

Definition

1. Let $\mathcal{R}$ be an equivalence relation on A and let $a \in A$. The set

$$[a] = \overline{x} = cl(x) = \{x \in A | a\mathcal{R}x\}$$

   is called the equivalence class of a.

2. The element in the bracket in the above notation is called **the Representative of the equivalence class.**

Example

1. $[0] = \mathbb{Z}$, the set of integers.
2. $[\frac{1}{2}] = \{\frac{m}{2} | m \ \ is \ \ an \ \ odd \ \ integer\}$
3. $[\pi] = \{\pi + n | n \ \ is \ \ an \ \ integer \ \} = [\pi + n]$, for any integer n.

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Theorem (Properties of equivalence classes)

Let $\mathcal{R}$ be an equivalence relation on a set A and let $a, b \in A$. Then

1. $a \in [a]$
2. *if* $b \in [a]$, *then* $[a] = [b]$
3. $[a] = [b]$ *iff* $(a, b) \in R$,
4. *if* $[a] \cap [b] \neq \emptyset$, *then* $[a] = [b]$.

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Proof:

1. By definition of $[a]$, we have

$$[a] = \{x \in A : (x, a) \in R\} = \{x \in A : xRa\}.$$

Since R is a reflexive relation, then

$$aRa, \forall a \in A.$$

Thus $a \in [a]$.

2. Suppose that $b \in [a]$ and let $x \in [b]$. Then $x \in [b] \Rightarrow (x, b) \in R$. Since $b \in [a]$, then $(b, a) \in R$. Hence $(x, b) \in R \wedge (b, a) \in R \Rightarrow (x, a) \in R$ (since R is transitive), $\Rightarrow x \in [a]$. Thus

$$[b] \subseteq [a]. \tag{1}$$

Now, suppose that $y \in [a]$. Then
$y \in [a] \Rightarrow (y, a) \in R$.
Hence

$$\begin{aligned} b \in [a] &\Rightarrow (b, a) \in R \\ &\Rightarrow (a, b) \in R, \end{aligned}$$

since R is symmetric. Consequently

$$\begin{aligned} (y, a) \in R \wedge (a, b) \in R &\Rightarrow (y, b) \in R \\ &\Rightarrow y \in [b], \end{aligned}$$

since R is transitive. Therefore

$$[a] \subseteq [b]. \tag{2}$$

Thus, it follows from Equations 1 and 2 that

$$[a] = [b].$$

3. Firstly, suppose that $[a] = [b]$. Then by part (1.), we have

$$a \in [a].$$

Since $[a] = [b]$, then

$$\begin{aligned} a \in [a] &\Rightarrow a \in [b] \\ &\Rightarrow (a, b) \in R. \end{aligned}$$

Conversely, suppose that $x \in [a]$. Then,
$x \in [a] \Rightarrow (x, a) \in R$. This implies $(x, a) \in R \wedge (a, b) \in R$.
Hence,

$$(x, a) \in R \wedge (a, b) \in R \quad \Rightarrow \quad (x, b) \in R \Rightarrow x \in [b].$$

Thus,

$$[a] \subseteq [b]. \tag{3}$$

By the same manner, we prove that

$$[b] \subseteq [a]. \tag{4}$$

From Equations 3and 4, we conclude

$$[a] = [b].$$

(4.) Suppose that $[a] \cap [b] \neq \emptyset$, and let $x \in [a] \cap [b]$. Then

$$\begin{aligned}
x \in [a] \cap [b] &\Rightarrow x \in [a] \wedge x \in [b] \\
&\Rightarrow (x, a) \in R \wedge (x, b) \in R \\
&\Rightarrow (x, b) \in R \wedge (x, a) \in R \\
&\Rightarrow (b, x) \in R \wedge (x, a) \in R(\ \textit{since}\ \ R\ \ \textit{is}\ \ \textit{symmetric}) \\
&\Rightarrow (b, a) \in R(\ \textit{since}\ \ R\ \ \textit{is}\ \ \textit{transitive}) \\
&\Rightarrow (a, b) \in R(\ \textit{since}\ \ R\ \ \textit{is}\ \ \textit{symmetric}) \\
&\Rightarrow [a] = [b](\textit{see}\ \ \textit{part}\ \ (3))
\end{aligned}$$

Partition

Definition

Let $\{A_i\}_{i\in I}$ be a family of nonempty subsets of A. Then $\{A_i\}_{i\in I}$ is called partition of A iff

1. $A_i \cap A_j = \emptyset \vee A_i = A_j$
2. $A = \bigcup A_i$.

Example

Let $A = \mathbb{Z}$ and let $X = \mathbb{Z}_e$ and $Y = \mathbb{Z}_o$. It is clear that X and Y are nonempty subsets of A and $X \cap Y = \emptyset$, also $A = X \cup Y$. Therefore, the set $\{X, Y\}$ is a partition of A.

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Theorem

Let R be an equivalence relation on a set A and let $\{[a]\}_{a\in A}$ be the family of all the equivalence classes of R in A. Then $\{[a]\}_{a\in A}$ is a partition of A.

Proof:

It is clear that $[a] \subseteq A$, $\forall a \in A$. Since R is a reflexive relation, then $(a, a) \in R$ and $a \in [a]$ (by Theorem "Properties of equivalence classes" part (1)). Thus $[a] \neq \emptyset$, $\forall a \in A$.

Now, suppose that $\exists a, b \in A$, such that $[a] \cap [b] \neq \emptyset$. Then $[a] = [b]$ (by Theorem "Properties of equivalence classes" part (4)). Finally, we must show that

$$A = \bigcup_{a \in A} [a].$$

Suppose that $x \in A$, then $x \in [x]$ (by Theorem "Properties of equivalence classes" part (1)). Thus, $x \in \bigcup_{a \in A}[a]$., that is $A \subseteq = \bigcup_{a \in A}[a]$. and it is clear that $= \bigcup_{a \in A}[a] \subseteq A$.

Thus,

$$A = \bigcup_{a \in A} [a].$$

Hence, $[a]_{a \in A}$ is a partition of A.

Theorem

If A is a nonempty set and $A_{i\,i\in I}$ is a partition of A, then there exists an equivalence relation R on A, such that $A_{i\,i\in I}$ are the equivalence classes of R.

Proof:

Let R be a relation on A defined by

$$R = \{(x, y) \in A \times A : \exists A_i, \ \textit{such that} \ \ x, y \in A_i\}.$$

Below we show that R is an equivalence relation.

1. Let $x \in A$. Since $A_{i\,i\in I}$ is a partition of A, then

$$A = \bigcup_{i\in I} A_i.$$

So, $\exists A_i$, such that $x \in A_i$. Then

$$x \in A_i \wedge x \in A_i \Rightarrow (x, x) \in R.$$

Thus R is a reflexive relation.
2. Suppose that $(x, y) \in R$. Then $\exists A_i$, such that $x, y \in A_i$. That is $\exists A_i, y \in A_i \wedge x \in A_i$. Hence $(y, x) \in R$. Hence R is symmetric.
3. Suppose that $(x, y) \in R \wedge (y, z) \in R$. Then $\exists A_i, A_j$, such that $x, y \in A_i \wedge y, z \in A_j$. Thus $y \in A_i \cap A_j$. Hence $A_i \cap A_j \neq \emptyset$, and $A_{i\,i\in I}$ is a partition of the set A.
Consequently $A_i = A_j$.
So, $\exists A_i$, such that $x \in A_i \wedge z \in A_i$ and from the definition of R, we have $(x, z) \in R$. Therefore, R is a transitive relation.
Thus R is an equivalence relation.

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Now, we show that every subset A_i in $A_{i\,i\in I}$ is an equivalence class for R. As $A_i \neq \emptyset$, $\forall i \in I$, then A_i has at least one element x, so $[x]$ is an equivalence class for R.
We claim that $A_i = [x]$. Suppose that $y \in [x]$. Then $(y, x) \in R$, and since $x \in A_i$, then by definition of R, $y \in A_i$. That is

$$y \in [x] \Rightarrow y \in A_i.$$

Then

$$\begin{aligned} & z \in A_i \wedge x \in A_i \\ & \Rightarrow (z, x) \in R \\ & \Rightarrow z \in [x]. \end{aligned}$$

Consequently,

$$z \in A_i \Rightarrow z \in [x],$$

that is $A_i \subseteq [x]$. Hence $A_i = [x]$.

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Definition

Let $\mathcal{R}$ be an equivalence relation on A. The quotient set of A modulo $\mathcal{R}$ is

$$A/\mathcal{R} = \{cl(a) | a \in A\}.$$

Example

Definition

1. Let S be a set and $\mathcal{R}$ a binary relation on S. The relation $\mathcal{R}$ is said to be **a partial order** (or poset) if it satisfies the following conditions:
   - $\mathcal{R}$ is re‡exive,
   - $\mathcal{R}$ is antisymmetric,
   - $\mathcal{R}$ is transitive.

   A set equipped with a partial order relation is called an ordered set. We often use $\leq$ to denote a partial order relation
2. A binary relation $\mathcal{R}$ on a set S is said to be a total order if:
   - $\mathcal{R}$ is a partial order,
   - For any $x, y \in S$, either xRy or yRx (i.e., any pair of elements of S is comparable under $\mathcal{R}$).

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Example

1. Let R_1 be a relation on $\mathbb{Z}$ defined by

$$S = \{(x, y) \in Z \times \mathbb{Z} : 3|(x - y)\},$$

R_1 is not a partial order relation on $\mathbb{Z}$, because R_1 is not antisymmetric. For example, $(8, 2) \in S \wedge (2, 8) \in S$, but $2 \neq 8$.

2. Let X be a nonempty set and R_2 be a relation on $P(X)$ defined by R_2 is a partial order relation on $P(X)$. then $(P(X), \subseteq)$ is a poset.

$$R = \{(A, B) \in P(X) \times P(X) : A \subseteq B\},$$

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Example

1. $(\mathbb{Z}, \leq)$ is a poset. Every pair of integers are related via $\leq$, so $\leq$ is a total order and $(\mathbb{Z}, \leq)$ is a chain.
2. $(\mathbb{Z}, |)$ is a poset. The relation $a|b$ means "a divides b."

Example

1. The division on $\mathbb{N}$, denoted $/$, is a partial order.
2. The relation less or equal to, denoted $\leq$, is a total order on $\mathbb{R}$.
3. The relation less than to, denoted $<$, is a strict order on $\mathbb{R}$.
4. The inclusion relation, denoted $\subset$ on the power set $P(S)$ of a set S is a partial order.
5. On $\mathbb{N}$, the relation $\mathcal{R}$, defined by: $a\mathcal{R}b$ if only if $b = 2^k a$ for some integer $k \geq 0$ is a partial order.